

PROJECT TITLE: Expense Tracker with Budget Alert

SUBJECT: DevOps

COURSE: BSc Semester 2

GROUP MEMBERS:

1. Raunak Kumar
 2. Hitarth Desai
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OBJECTIVE:

The objective of this project is to track daily expenses and compare them with a monthly budget using a Python-based system. The system helps users manage their spending effectively.

```
Raunak@DESKTOP-I6E0E1U MINGW64 ~/Desktop/Expense_tracker (main)
$ python --version
Python 3.14.4
```

FEATURES:

1. Add Expense
 2. View All Expenses
 3. Show Category-wise Total
 4. Set Monthly Budget
 5. Alert if Budget Crossed
 6. Exit
-

SYSTEM FLOW:

The program starts and displays a menu.

The user selects an option from the menu.

Based on the selected option, the system performs operations such as adding expenses, viewing records, calculating totals, or checking the budget.

All expenses are stored in a list structure.

The system processes data and displays results accordingly.

LOGIC USED:

- A list is used to store all expenses.
 - A tuple is used to store category and amount.
 - A dictionary is used to calculate category-wise totals.
 - A loop is used to continuously display the menu.
 - Conditional statements are used for decision making.
-

DEVOPS LIFECYCLE:

Plan: Define project features and structure

Code: Implement Python program

Build: Prepare requirements file and environment

Test: Perform manual testing of features

Deploy: Run application using Docker container

TIMELINE:

Day 1 – Planning and requirement analysis

Day 2 – Coding and implementation

Day 3 – Testing and debugging

Day 4 – Git version control

Day 5 – Docker deployment

SOURCE CODE (main.py):

```
print("Expense Tracker Starting...")
```

```
expenses = []
```

```
budget = 0
```

```
def add_expense():
```

```
    category = input("Enter category: ")
```

```
    try:
```

```
        amount = float(input("Enter amount: "))
```

```
if amount <= 0:
    print("Invalid Amount")
else:
    expenses.append((category, amount))
    print("Expense Added Successfully")
```

```
except:
    print("Invalid Input")
```

```
def view_expenses():
    if len(expenses) == 0:
        print("No expenses found")
    else:
        for item in expenses:
            print(item[0], item[1])
```

```
def category_total():
    totals = {}

    for item in expenses:
        cat = item[0]
        amt = item[1]

        if cat in totals:
            totals[cat] += amt
        else:
            totals[cat] = amt
```

```
for key in totals:  
    print(key, totals[key])
```

```
def set_budget():  
    global budget  
    budget = float(input("Enter budget: "))  
    print("Budget Set")
```

```
def check_budget():  
    total = 0
```

```
for item in expenses:  
    total += item[1]
```

```
print("Total Expense:", total)
```

```
if total > budget:  
    print("ALERT: Budget Crossed!")  
else:  
    print("Within Budget")
```

```
def show_menu():  
    print("\n--- Expense Tracker Menu ---")  
    print("1. Add Expense")  
    print("2. View All Expenses")  
    print("3. Show Category-wise Total")  
    print("4. Set Monthly Budget")  
    print("5. Check Budget Alert")
```

```
print("6. Exit")
```

```
while True:
```

```
    show_menu()
```

```
    choice = input("Enter choice: ")
```

```
    if choice == "1":
```

```
        add_expense()
```

```
    if choice == "2":
```

```
        view_expenses()
```

```
    if choice == "3":
```

```
        category_total()
```

```
    if choice == "4":
```

```
        set_budget()
```

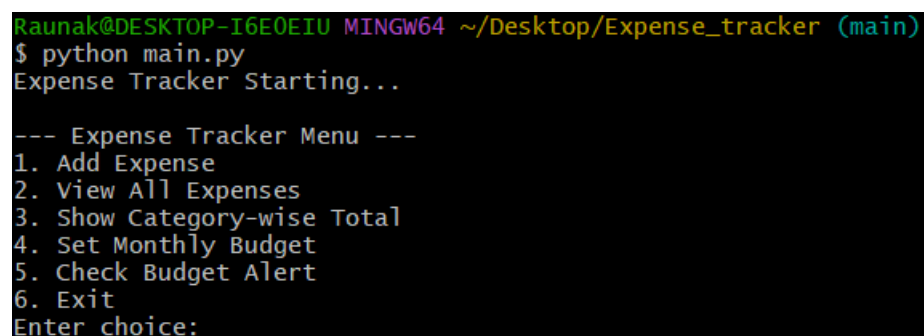
```
    if choice == "5":
```

```
        check_budget()
```

```
    if choice == "6":
```

```
        print("Thank You")
```

```
    break
```



```
Raunak@DESKTOP-I6E0EIU MINGW64 ~/Desktop/Expense_tracker (main)
$ python main.py
Expense Tracker Starting...

--- Expense Tracker Menu ---
1. Add Expense
2. View All Expenses
3. Show Category-wise Total
4. Set Monthly Budget
5. Check Budget Alert
6. Exit
Enter choice:
```

GIT IMPLEMENTATION:

Git was used for version control of the project. Multiple commits were made to track the development process step-by-step.

```
Raunak@DESKTOP-I6E0EIU MINGW64 ~/Desktop/Expense_tracker (main)
$ git log --oneline
10850ea (HEAD -> main, origin/main, origin/HEAD) Rename DockerFile to Dockerfile
7d4adfa DockerFile Added
d3cbc08 Added features like Set budget and Check budget alert
3366a33 Added view expenses feature and category total function
c77f765 Added add expense feature
567402e Added main menu and loop
3a2aa1d Initial Commit
```

TESTING REPORT:

Test Case | Input | Expected Result | Actual Result | Status

Add Expense	Food, 200	Expense Added Successfully	Expense Added Successfully	Pass
View Expenses	-	Display list of expenses	Display list of expenses	Pass
Category Total	Food	Show total for category	Show total for category	Pass
Set Budget	5000	Budget set successfully	Budget set successfully	Pass
Budget Alert	>5000	Alert shown	Alert shown	Pass

DOCKER DEPLOYMENT:

The application was containerized using Docker. A Dockerfile was created to define the execution environment. The Docker image was successfully built and the application was executed inside a container.

```

PS C:\Users\Admin\Desktop\Expense_tracker> docker build -t expenseapp .
[+] Building 392.9s (9/9) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 112B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11     4.4s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/3] FROM docker.io/library/python:3.11@sha256:5062f6c22a2c3b8b3717c642af3852310d9a81c3c2c0fc72449a9a79ef0 387.7s
=> => resolve docker.io/library/python:3.11@sha256:5062f6c22a2c3b8b3717c642af3852310d9a81c3c2c0fc72449a9a79ef09a 0.0s
=> => sha256:c380448934e9e89b6e97e314f2d21c13d73788207dc050206abe91bccfcf38e6 249B / 249B 0.8s
=> => sha256:6cf369abc841afa47f3a93567981f9c1d723f382d7fb9db98b8099bf8b63c5 23.98MB / 23.98MB 75.5s
=> => sha256:5e69c8985c08e44d57de4d2a9908b23fe70ae893e35e66e78afc489f0a6b0f9d 6.09MB / 6.09MB 26.8s
=> => sha256:1e1bb20756a85e159dfb7908a2a5c860ac2f2254254a915d8c047e02cb2b0259 236.07MB / 236.07MB 384.0s
=> => sha256:20d12c8f3f3fbb5bd2b8369adf3213e09d6b33f975462543ddfece85b2fe85e5 67.79MB / 67.79MB 228.6s
=> => sha256:5467f93954cfe1451f433422086d00bd066cf33f42112b531804ffe06f0a929 25.62MB / 25.62MB 98.8s
=> => sha256:3b32e3bb7338c216b077e95920ae53332d81bd57f4a7393bc324432d5a3f88a2 49.30MB / 49.30MB 168.8s
=> => extracting sha256:3b32e3bb7338c216b077e95920ae53332d81bd57f4a7393bc324432d5a3f88a2 0.7s
=> => extracting sha256:5467f93954cfe1451f433422086d00bd066cf33f42112b531804ffe06f0a929 0.4s
=> => extracting sha256:20d12c8f3f3fbb5bd2b8369adf3213e09d6b33f975462543ddfece85b2fe85e5 1.0s
=> => extracting sha256:1e1bb20756a85e159dfb7908a2a5c860ac2f2254254a915d8c047e02cb2b0259 2.9s
=> => extracting sha256:5e69c8985c08e44d57de4d2a9908b23fe70ae893e35e66e78afc489f0a6b0f9d 0.1s
=> => extracting sha256:6cf369abc841afa47f3a93567981f9c1d723f382d7fb9db98b8099bf8b63c5 0.4s
=> => extracting sha256:c380448934e9e89b6e97e314f2d21c13d73788207dc050206abe91bccfcf38e6 0.0s
=> [internal] load build context                                   0.2s
=> => transferring context: 38.46kB                                  0.1s
=> [2/3] WORKDIR /app                                           0.2s
=> [3/3] COPY . .                                               0.0s
=> exporting to image                                           0.3s
=> => exporting layers                                             0.1s
=> => exporting manifest sha256:fa263328cb93c5c5b0b2f75ce26e04eebb457a8c94f5dfa00a627a9edf837579 0.0s
=> => exporting config sha256:d9293abb11a90ac6631580ab07a98cbc004f02715f112f23a6b66c4d0fd6e5d 0.0s
=> => exporting attestation manifest sha256:c9aaf3334f2422322d83497b8c51610e98802aa0c8a1406d56a0f7cdb0dcbf 0.0s
=> => exporting manifest list sha256:98ebad2fa4cfbf8d2dd6ec83f4ed1c82963b75753c871606c8734940511c6f6c 0.0s
=> => naming to docker.io/library/expenseapp:latest              0.0s
=> => unpacking to docker.io/library/expenseapp:latest           0.0s

```

```

PS C:\Users\Admin\Desktop\Expense_tracker> docker run -it expenseapp
Expense Tracker Starting...

```

```

--- Expense Tracker Menu ---
1. Add Expense
2. View All Expenses
3. Show Category-wise Total
4. Set Monthly Budget
5. Check Budget Alert
6. Exit
Enter choice: 6
Thank You

```

TEAM CONTRIBUTION:

Raunak Kumar:

- Project planning
- Python coding and logic implementation
- Core functionality development
- Docker deployment

Hitarth Desai:

- GitHub repository management
- Testing of application
- Documentation and formatting